

KEMIJARVI, Finland

WMO: 028140

Lat: 66.716N

Lon: 27.159E

Elev: 208

StdP: 98.85

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.4	(c) -28.3	(d) -34.5	(e) 0.1	(f) -31.4	(g) -31.0	(h) 0.2	(i) -28.2	(j) 10.1	(k) -4.7	(l) 9.1	(m) -4.1	(n) 1.1	(o) 320	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	25.3	16.1	23.1	15.5	21.1	14.5	18.0	22.6	16.8	21.2	15.7	19.4	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.0	19.5	15.2	11.0	18.1	14.1	10.3	17.1	51.7	22.6	47.9	21.1	44.5	19.4	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.3	(b) 7.3	(c) 6.5	(e) -34.9	(f) 27.1	(g) 2.4	(h) 2.7	(i) -36.6	(j) 29.0	(k) -38.0	(l) 30.6	(m) -39.4	(n) 32.2	(o) -41.1	(p) 34.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-13.0	-11.1	-7.2	-1.1	6.2	11.1	14.8	12.2	7.8	0.3	-4.6	-8.4		
	DBStd	10.65	7.37	7.94	5.76	4.34	4.73	4.14	3.51	3.50	3.33	4.21	5.34	7.04		
	HDD10.0	3756	712	592	534	333	138	35	4	18	81	300	438	571		
	HDD18.3	6472	970	825	793	583	376	220	118	194	317	558	688	829		
	CDD10.0	339	0	0	0	0	20	68	153	84	14	0	0	0		
	CDD18.3	14	0	0	0	0	0	3	9	2	0	0	0	0		
	CDH23.3	165	0	0	0	0	8	33	102	22	0	0	0	0		
	CDH26.7	25	0	0	0	0	0	5	16	4	0	0	0	0		
(14) Wind	WSAvg	2.9	2.6	2.9	3.3	2.9	3.2	3.2	2.7	3.0	2.9	2.8	2.8	3.0		
(15) Precipitation	PrecAvg	581	43	35	32	29	47	61	76	61	55	51	46	45		
	PrecMax	793	73	70	53	64	96	99	112	138	112	91	78	76		
	PrecMin	449	17	6	10	11	14	15	21	28	8	20	15	15		
	PrecStd	97	17	19	13	15	19	26	27	29	28	21	18	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	2.9	5.2	13.5	24.5	27.0	29.0	26.6	18.2	10.1	5.1	2.7		
		MCWB	0.1	1.9	2.1	6.9	12.9	16.4	18.1	19.0	13.5	9.3	4.7	2.2		
	2%	DB	-0.4	1.0	2.9	8.7	21.5	23.5	26.1	22.5	16.3	8.2	3.6	1.3		
		MCWB	-0.7	0.5	0.7	4.3	12.6	15.6	16.9	16.0	12.4	7.0	3.2	1.0		
	5%	DB	-1.5	0.1	1.4	6.6	17.6	20.8	23.8	20.4	14.8	7.1	2.3	0.5		
		MCWB	-1.8	-0.3	-0.1	3.0	10.9	13.3	16.3	15.0	11.6	6.2	2.1	0.2		
	10%	DB	-3.0	-1.1	0.3	5.0	14.5	18.5	22.0	18.3	13.3	5.9	1.3	-0.2		
		MCWB	-3.2	-1.5	-0.7	2.0	9.1	12.5	15.6	14.0	11.1	5.2	1.0	-0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	2.1	2.5	7.4	14.2	18.1	20.1	19.3	14.9	9.7	5.0	2.2		
		MCDB	0.4	2.9	4.4	12.6	21.5	23.2	24.3	24.7	16.6	10.1	5.0	2.5		
	2%	WB	-0.6	0.6	1.1	5.0	12.9	16.8	18.6	17.2	13.1	7.2	3.2	1.0		
		MCDB	-0.5	0.9	2.3	8.2	20.3	22.7	22.5	21.2	15.2	7.8	3.4	1.2		
	5%	WB	-1.7	-0.2	0.3	3.4	11.5	15.0	17.4	15.8	12.2	6.2	2.1	0.3		
		MCDB	-1.5	0.1	1.1	5.7	17.2	18.9	21.7	18.4	14.0	6.9	2.4	0.5		
	10%	WB	-3.2	-1.5	-0.5	2.4	9.6	13.4	16.3	14.8	11.4	5.3	1.1	-0.4		
		MCDB	-3.0	-1.1	0.3	4.3	13.7	16.9	20.5	17.3	13.0	5.8	1.3	-0.2		
(35) Mean Daily Temperature Range	MDBR		7.7	7.6	9.7	8.8	9.1	8.4	8.9	8.5	7.0	4.8	5.3	6.5		
	5% DB	MCDBR	6.9	5.5	7.7	10.7	14.9	12.4	12.9	12.0	10.6	5.7	4.1	4.7		
		MCWBR	7.0	5.2	6.2	9.7	7.4	5.7	5.4	6.1	7.0	4.8	4.1	4.6		
	5% WB	MCDBR	7.2	5.4	5.6	9.4	13.5	10.3	10.8	9.2	8.0	5.2	4.0	4.7		
		MCWBR	7.2	5.2	4.8	6.2	7.2	5.4	5.6	5.3	5.7	4.7	4.0	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.194	0.261	0.294	0.327	0.340	0.342	0.355	0.338	0.313	0.283	0.350	N/A		
	taud		1.926	2.099	2.160	2.226	2.443	2.475	2.454	2.483	2.535	2.339	2.687	N/A		
	Ebn at Noon		193	572	752	827	861	868	840	820	759	563	212	0		
	Edh at Noon		15	53	84	104	94	93	93	81	61	42	16	0		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.57	2.05	3.82	4.77	5.12	4.75	3.46	1.80	0.61	0.12	0.00		
	RadStd		0.00	0.10	0.20	0.35	0.41	0.38	0.47	0.33	0.17	0.12	0.02	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page